

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

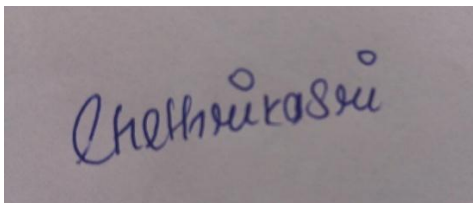
Total Marks: 50

Annexure A

Code of Conduct:

I ___chethrikasri km_(192511112)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a handwritten signature in blue ink on a light-colored surface. The signature appears to be 'Chethrikasri' written in a cursive style.

Question 1: RSA Key Generation and Security Analysis

Problem Statement

Given:

- Prime numbers: $p = 17$, $q = 19$
- Public exponent: $e = 5$
- Plaintext message: $M = 7$

Required:

- Compute n and $\phi(n)$.
- Calculate the private key d .
- Encrypt the message $M = 7$.
- Decrypt the ciphertext and verify the result.
- Analyze the impact of using small prime numbers on RSA security.

Introduction

RSA (Rivest–Shamir–Adleman) is one of the most widely used public-key cryptographic algorithms for secure communication. It uses two different keys: a public key for encryption and a private key for decryption. The public key is shared with everyone, while the private key is kept secret by the receiver. The security of RSA depends on the difficulty of factoring the product of two large prime numbers. This makes it suitable for secure communication, digital signatures, authentication, and key exchange.

Given Data

Parameter	Value
Prime Number (p)	17
Prime Number (q)	19
Public Exponent (e)	5
Plaintext Message (M)	7

Solution

Step 1: Calculate the Modulus (n)

The modulus is calculated by multiplying the two prime numbers.

Formula

$$n = p \times q$$

Calculation

$$n = 17 \times 19$$

$$n = 323$$

Result

$$n = 323$$

Explanation

The modulus n is an important parameter in the RSA algorithm. It is included in both the public key and the private key and is used during both encryption and decryption.

Step 2: Calculate Euler's Totient Function

Euler's Totient Function is calculated using the following formula.

Formula

Calculation

$$\phi(n) = (p - 1) \times (q - 1)$$

$$\phi(n) = (17 - 1) \times (19 - 1)$$

$$\phi(n) = 16 \times 18$$

$$\phi(n) = 288$$

Result

$$\phi(n) = 288$$

Explanation

The value of $\phi(n)$ is used to generate the private key. It represents the number of integers that are relatively prime to the modulus n .

Step 3: Calculate the Private Key (d)

The public exponent is given as

$$e = 5$$

The private key must satisfy the following condition.

$$5 \times d \equiv 1 \pmod{288}$$

Using the Extended Euclidean Algorithm,

$$d = 173$$

Verification

$$5 \times 173 = 865$$

$$865 = (3 \times 288) + 1$$

Therefore,

$$865 \bmod 288 = 1$$

Hence,

$$\text{Private Key (d)} = 173$$

The generated keys are:

$$\text{Public Key} = (5, 323)$$

$$\text{Private Key} = (173, 323)$$

Explanation

The public key is used for encryption, while the private key is used only by the receiver to decrypt the ciphertext.

Step 4: Encrypt the Message

Encryption is performed using the public key.

Encryption Formula

$$C = M^e \bmod n$$

Substituting the values

$$C = 7^5 \bmod 323$$

$$7^5 = 16807$$

$$16807 \bmod 323 = 11$$

Result

$$\text{Ciphertext (C)} = 11$$

Explanation

The plaintext message is converted into ciphertext using the public key. The encrypted message cannot be understood without the corresponding private key.

Step 5: Decrypt the Ciphertext

Decryption is performed using the private key.

Decryption Formula

$$M = C^d \bmod n$$

Substituting the values

$$M = 11^{173} \bmod 323$$

Using modular exponentiation,

$$M = 7$$

Result

Recovered Plaintext = 7

Verification

Original Message = 7

Decrypted Message = 7

Since both values are the same, the RSA encryption and decryption process is verified successfully.

Security Analysis

The security of RSA depends on the difficulty of factoring the modulus into its prime factors. In this problem,

$$323 = 17 \times 19$$

Since the prime numbers are very small, the modulus can be factorized easily. Once an attacker determines the values of p and q , they can calculate Euler's Totient Function, derive the private key, and decrypt the encrypted message.

In practical applications, RSA uses prime numbers that are 2048 bits or larger. Large prime numbers make factorization extremely difficult, thereby providing strong security against brute-force and factorization attacks.

Impact of Using Small Prime Numbers

- Easy to factor the modulus.
- Private key can be recovered quickly.
- Weak security against attackers.
- Not suitable for real-world secure communication.
- Useful only for learning and demonstration purposes.

Conclusion

The RSA algorithm was successfully implemented by generating the public and private keys, encrypting the plaintext message, and decrypting the ciphertext back to its original form. The results confirm that the RSA algorithm works correctly. The security analysis also shows that using small prime numbers weakens RSA security because the modulus can be factorized easily. Therefore, large prime numbers should always be used in practical cryptographic applications.

Final Result

Parameter	Result
Modulus (n)	323
Euler's Totient Function $\phi(n)$	288
Public Key	(5, 323)
Private Key	173
Plaintext Message	7
Ciphertext	11
Decrypted Message	7
Verification	Successful

Question 2: ElGamal Encryption Analysis

Problem Statement

Given:

- Prime Number: $p = 23$
- Generator: $g = 5$
- Private Key: $x = 7$
- Random Number: $k = 3$
- Plaintext Message: $M = 10$

Required:

- Compute the public key.
- Perform encryption to generate the ciphertext pair (C_1, C_2) .
- Decrypt the ciphertext to retrieve the original message.
- Analyze how the random value k improves the security of ElGamal encryption.

Introduction

ElGamal is a public-key cryptographic algorithm based on the Discrete Logarithm Problem (DLP). It was proposed by Taher ElGamal in 1985 and is widely used for secure communication and digital signatures. Unlike RSA, ElGamal introduces a random value k during encryption, ensuring that the same plaintext produces different ciphertexts each time it is encrypted. This randomness makes ElGamal highly secure against several cryptographic attacks.

Given Data

Parameter	Value
Prime Number (p)	23
Generator (g)	5
Private Key (x)	7
Random Number (k)	3
Plaintext Message (M)	10

Solution

Step 1: Compute the Public Key

The public key is calculated using the formula:

Formula

$$y = g^x \bmod p$$

Calculation

$$y = 5^7 \bmod 23$$

$$5^7 = 78125$$

$$78125 \bmod 23 = 17$$

Result

Public Key (y) = 17

Therefore, the public key is:

$(p, g, y) = (23, 5, 17)$

Step 2: Compute the First Ciphertext Component (C_1)

The first ciphertext component is calculated using:

Formula

$$C_1 = g^k \bmod p$$

Calculation

$$C_1 = 5^3 \bmod 23$$

$$5^3 = 125$$

$$125 \bmod 23 = 10$$

Result

$$C_1 = 10$$

Explanation

The value of C_1 depends on the random number k . Every time a different value of k is selected, a different ciphertext is produced.

Step 3: Compute the Second Ciphertext Component (C_2)

The second ciphertext component is calculated using:

Formula

$$C_2 = M \times y^k \bmod p$$

First calculate:

$$17^3 \bmod 23$$

$$17^2 = 289$$

$$289 \bmod 23 = 13$$

$$17^3 = 13 \times 17 = 221$$

$$221 \bmod 23 = 14$$

Now,

$$C_2 = 10 \times 14 \bmod 23$$

$$C_2 = 140 \bmod 23$$

$$C_2 = 2$$

Result

$$C_2 = 2$$

Therefore, the ciphertext pair is

$$(C_1, C_2) = (10, 2)$$

Explanation

The ciphertext consists of two values, C_1 and C_2 , both of which are required to recover the original plaintext during decryption.

Step 4: Decrypt the Ciphertext

The receiver first computes the shared secret.

Formula

$$S = C_1^x \bmod p$$

Calculation

$$S = 10^7 \bmod 23$$

$$S = 14$$

Now find the modular inverse of **14** modulo **23**.

$$14^{-1} \bmod 23 = 5$$

Recover the plaintext using

$$M = C_2 \times 14^{-1} \bmod 23$$

$$M = 2 \times 5 \bmod 23$$

$$M = 10 \bmod 23$$

$$M = 10$$

Result

Recovered Plaintext = 10

Verification

Original Message = 10

Recovered Message = 10

Since both values are equal, the encryption and decryption process is verified successfully.

Security Analysis

The security of the ElGamal cryptosystem depends on the Discrete Logarithm Problem, which is computationally difficult to solve for large prime numbers.

A unique feature of ElGamal is the use of the random value **k** during encryption. Even if the same plaintext message is encrypted multiple times using the same public key, different values of **k** generate different ciphertexts. This property prevents attackers from identifying repeated messages and provides semantic security.

Role of Randomness (k)

- Produces different ciphertexts for the same plaintext.
- Prevents pattern analysis attacks.
- Enhances confidentiality of transmitted data.
- Makes brute-force attacks more difficult.
- Improves overall security of the encryption process.

Conclusion

The ElGamal encryption algorithm was successfully implemented by generating the public key, encrypting the plaintext message, and decrypting the ciphertext to recover the original message.

Final Result

Parameter	Result
Prime Number (p)	23
Generator (g)	5
Private Key (x)	7
Public Key (y)	17
Random Number (k)	3
Ciphertext (C ₁ , C ₂)	(10, 2)
Original Message	10
Decrypted Message	10
Verification	Successful

Question 3: RSA Digital Signature Verification

Problem Statement

Given:

- Prime numbers: $p = 11$, $q = 17$
- Public exponent: $e = 7$
- Message: $M = 8$

Required:

- Compute n and $\phi(n)$.
- Find the private key d .
- Generate the digital signature.
- Verify the signature using the public key.
- Explain how digital signatures ensure non-repudiation.

Introduction

RSA Digital Signature is used to verify the authenticity and integrity of a message. The sender signs the message using the private key, and the receiver verifies it using the public key. This ensures that the message has not been altered and confirms the sender's identity.

Given Data

Parameter	Value
Prime Number (p)	11
Prime Number (q)	17
Public Exponent (e)	7
Message (M)	8

Solution

Step 1: Calculate n and $\phi(n)$

Formula

$$n = p \times q$$

$$\phi(n) = (p - 1)(q - 1)$$

Calculation

$$n = 11 \times 17 = 187$$

$$\phi(n) = (10 \times 16) = 160$$

Step 2: Calculate the Private Key

The private key satisfies:

$$7 \times d \equiv 1 \pmod{160}$$

Using the Extended Euclidean Algorithm,

$$d = 23$$

Verification

$$7 \times 23 = 161$$

$$161 \bmod 160 = 1$$

Therefore,

Private Key = 23

Public Key = (7, 187)

Private Key = (23, 187)

Step 3: Generate the Digital Signature

The signature is generated using the private key.

Formula

$$S = M^d \bmod n$$

Calculation

$$S = 8^{23} \bmod 187$$

Using modular exponentiation,

$$S = 162$$

Therefore,

$$\text{Digital Signature} = 162$$

Step 4: Verify the Signature

The receiver verifies the signature using the public key.

Formula

$$M = S^e \bmod n$$

Calculation

$$M = 162^7 \bmod 187$$

$$M = 8$$

Since the recovered message matches the original message, the signature is valid.

Security Analysis

RSA digital signatures provide authentication, integrity, and non-repudiation. Since only the sender possesses the private key, no one else can generate the same signature. If the message is modified during transmission, the verification process will fail. Therefore, digital signatures confirm both the sender's identity and the originality of the message.

Conclusion

The RSA digital signature was successfully generated and verified. The verified message matched the original message, proving that the message was authentic and had not been modified during transmission.

Final Result

Parameter	Result
n	187
$\phi(n)$	160
Private Key (d)	23
Digital Signature	162
Verified Message	8
Verification	Successful

Question 4: Diffie–Hellman with Attack Resistance Evaluation

Problem Statement

Given:

- Prime Number: $p = 29$
- Generator: $g = 2$
- User A Private Key = 5
- User B Private Key = 12

Required:

- Compute the public keys.
- Compute the shared secret key.
- Analyze the effect of a public key replacement attack.
- Suggest a solution to ensure authenticity.

Introduction

The Diffie–Hellman algorithm is a key exchange technique that allows two users to securely establish a shared secret key over an insecure network. The generated key is then used for secure communication.

Given Data

Parameter	Value
Prime Number (p)	29
Generator (g)	2
User A Private Key	5
User B Private Key	12

Solution

Step 1: Compute Public Keys

Formula

Public Key = $g^x \bmod p$

User A

$$A = 2^5 \bmod 29 = 3$$

User B

$$B = 2^{12} \bmod 29 = 7$$

Step 2: Compute the Shared Secret Key

For User A:

$$K = 7^5 \bmod 29 = 16$$

For User B:

$$K = 3^{12} \bmod 29 = 16$$

Shared Secret Key = 16

Attack Analysis

If an attacker intercepts and replaces the public keys, both users may unknowingly establish separate secret keys with the attacker instead of each other. This is known as a Man-in-the-Middle (MITM) attack, allowing the attacker to read or modify the communication.

Suggested Solution

To prevent this attack, Diffie–Hellman should be combined with Digital Signatures or Digital Certificates. These methods verify the identity of both users and ensure that the exchanged public keys are authentic.

Conclusion

The Diffie–Hellman algorithm successfully generated the same shared secret key (**16**) for both users. However, without authentication, it is vulnerable to Man-in-the-Middle attacks. Using digital signatures or certificates improves the security of the key exchange.

Final Result

Parameter	Result
User A Public Key	3
User B Public Key	7
Shared Secret Key	16
Attack	Man-in-the-Middle (MITM)
Suggested Solution	Digital Signatures / Digital Certificates

Question 5: Hybrid Encryption System Analysis

Problem Statement

Given:

- RSA is used for key exchange.
- AES is used for data encryption.
- RSA encryption time = 5 ms per key exchange.
- AES encryption time = 0.5 ms per data block.
- Total data blocks = 500.

Required:

- Calculate the total encryption time.
- Compare it with using only RSA.
- Explain why hybrid encryption is more efficient.

Introduction

A hybrid encryption system combines the advantages of RSA and AES. RSA is used to securely exchange the encryption key, while AES is used to encrypt the actual data. This approach provides both security and high performance.

Given Data

Parameter	Value
RSA Time	5 ms
AES Time per Block	0.5 ms
Total Data Blocks	500

Solution

Step 1: Calculate AES Encryption Time

AES Time = Number of Blocks $\times$ Time per Block

AES Time = 500×0.5

AES Time = 250 ms

Step 2: Calculate Total Hybrid Encryption Time

Total Time = RSA Time + AES Time

Total Time = $5 + 250$

Total Hybrid Encryption Time = 255 ms

Step 3: Compare with RSA Only

If only RSA is used for all 500 blocks,

RSA Time = 500×5

RSA Time = 2500 ms

Thus,

- Hybrid Encryption = 255 ms
- RSA Only = 2500 ms

Hybrid encryption is much faster than using RSA alone.

Analysis

Hybrid encryption is more efficient because RSA is computationally expensive and is used only once for key exchange. AES is much faster and encrypts large amounts of data quickly. This combination provides both high security and better performance.

Conclusion

The hybrid encryption system requires only 255 ms, whereas using only RSA requires 2500 ms. Therefore, hybrid encryption is more efficient and is widely used in secure communication systems such as HTTPS, VPNs, and secure file transfer.

Final Result

Parameter	Result
AES Encryption Time	250 ms
RSA Key Exchange Time	5 ms
Total Hybrid Encryption Time	255 ms
RSA Only Encryption Time	2500 ms
More Efficient Method	Hybrid Encryption

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation